

- 1 (a) Use standard results from the list of formulae (MF19) to find $\sum_{r=1}^n (3r^2 + 3r + 1)$ in terms of n , simplifying your answer. [3]

[illegible]

$$\frac{1}{r^3} - \frac{1}{(r+1)^3} = \frac{3r^2 + 3r + 1}{r^3(r+1)^3}$$

and hence use the method of differences to find $\sum_{r=1}^n \frac{3r^2+3r+1}{r^3(r+1)^3}$. [5]

This image shows a full page of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(c) Deduce the value of $\sum_{r=1}^{\infty} \frac{3r^2 + 3r + 1}{r^3(r+1)^3}$. [1]

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